ASSIGNMENT 11

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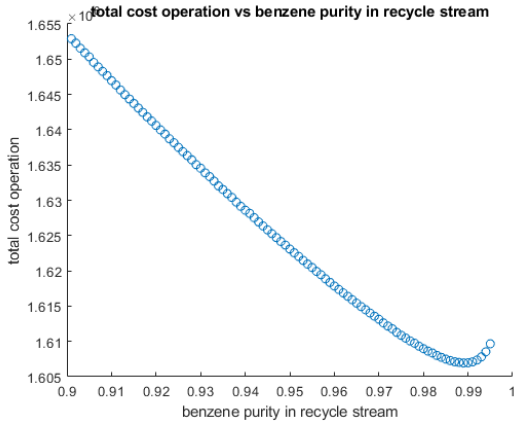
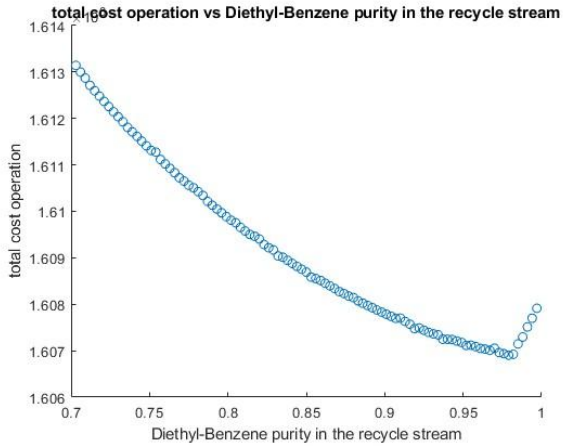
221036

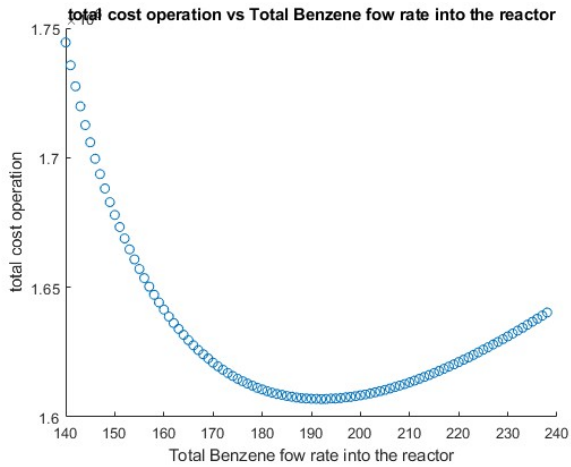
a) Total cost operation vs benzene purity in recycle stream



b) Total cost operation vs Diethyl-Benzene purity in the recycle stream



c) Total cost operation vs Total benzene flow rate into the reactor



FMINCON:

Iter	F-count	f(x)	Feasibility	First-order optimality	Norm of step
1	16	1.607004e+06	0.000e+00	5.704e+03	1.082e-01
2	21	1.606892e+06	0.000e+00	3.125e+04	1.195e-03
3	35	1.606860e+06	0.000e+00	1.170e+04	3.375e-03
4	50	1.606858e+06	0.000e+00	1.048e+04	1.849e-04
5	64	1.606857e+06	0.000e+00	1.153e+04	2.023e-05
6	76	1.606857e+06	0.000e+00	5.905e+04	8.849e-06
7	88	1.606857e+06	0.000e+00	1.135e+04	3.865e-06
8	99	1.606857e+06	0.000e+00	3.175e+07	3.388e-06
9	106	1.606857e+06	0.000e+00	3.175e+07	5.485e-08

Optimisation:

Benzene mole fraction in C2 distillate :0.9879

Diethyl-Benzene purity in the recycle stream : 0.0193

Total Benzene flow rate into the reactor: 189.6955

Optimised value: 1606900

Original TAC value:

```
>> TAC([0.987 0.02 190])

ans =

    1.6070e+06

fx >>
```